

EE 507: Transmission Lines and Basic Microwave Engineering

Course Code : EE 507

Course Name : Transmission Lines and Basic Microwave Engineering

Course L-T-P-C : 3-1-0-4

Prerequisite : Consent of the faculty member

Intended for : UG/PG

Distribution : Discipline Elective for 3rd and 4th year BTech in EE, MTech in EE and MS & PhD in the area

Approval: 12th Senate

Course Contents

- **Introduction to transmission lines** : Basic Transmission line equations: Two line theory, capacitance and inductance in transmission lines and impedance of loaded and unloaded transmission line. Reflections and VSWR: Scattering matrix and impedance transformation. Smith chart: Stub matching. [10 Lectures]
- **Guided EM waves**: - Wave propagation in different types of transmission lines: Co-axial, microstrip, strip-lines, co-planar lines and co-planar wave guides lines. Loss and loss-less transmission lines. Metallic and dielectric wave guides: Propagation of modes. [11 Lectures]
- **Introduction to Left Handed medium**: - Wave Propagation in Left-Handed medium: metamaterial-transmission line approach. Introduction to Electromagnetic Band-gap (EBG) structures. [8 Lectures]
- **Basic microwave devices**: - H-Plane, E-Plane and Magic-T, Wave-Guide Couplers. Ferrites & ferrite devices. Diodes: Schottky, PIN, Varactor. Bipolar Junction Transistors (BJT). [10 Lectures]
- **Microwave Resonators**: - Series and parallel resonators, Transmission line resonators, Dielectric resonators and its applications. Microwave cavity: Klystron, Magnetron, TWT. Excitation techniques of resonators. [10 Lectures]
- **Microwave propagation**: - Effects of atmosphere and ground on microwave propagation, plasma Effects, microwave heating. Introduction to wire-less power transfer. Biological effects of microwave radiation and safety. [7 Lectures]

Note: All the Units will have tutorials and in the end of course there will be a presentation assignment in which student (or a group) will present latest advancement in any-one of the topics taught in class

Text Books:

1. R K Shevgaonkar, **Electromagnetic Waves**, McGraw Hill Education, India, 2006.
2. David M Pozar, **Microwave Engineering**, 4th Edition, John Wiley & Sons.

References:

1. C. Caloz and T. Itho, **Electromagnetic Metamaterial: Transmission Line Theory and Microwave Applications**, Wileys Publications.
2. Research Papers as instructed by course Instructors.